

HAT-P-3b

R-1255

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-3_210429 · created 2026-07-31T12:12:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459333.902 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.018
Transit depth (Rp/Rs)^2	2.01 +/- 0.51 [%]
Orbital Inclination (inc)	87.55 +/- 1.15
Airmass coefficient 1 (a1)	1.012 +/- 0.022
Airmass coefficient 2 (a2)	-0.0068 +/- 0.0093
Scatter in the residuals of the lightcurve fit is	2.15 %
Best Comparison Star	#2 - [496, 185]
Optimal Aperture	3.9
Optimal Annulus	10.74
Transit Duration (day)	0.0913 +/- 0.0099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.018 vs 0.1063 (deviation 1.79σ)
Duration fit vs published	0.0913 d vs 0.0867 d (deviation 0.42σ)
Mid-time O–C	6.5 min
Depth SNR	7.1
Residual scatter	2.15 %

Light curve

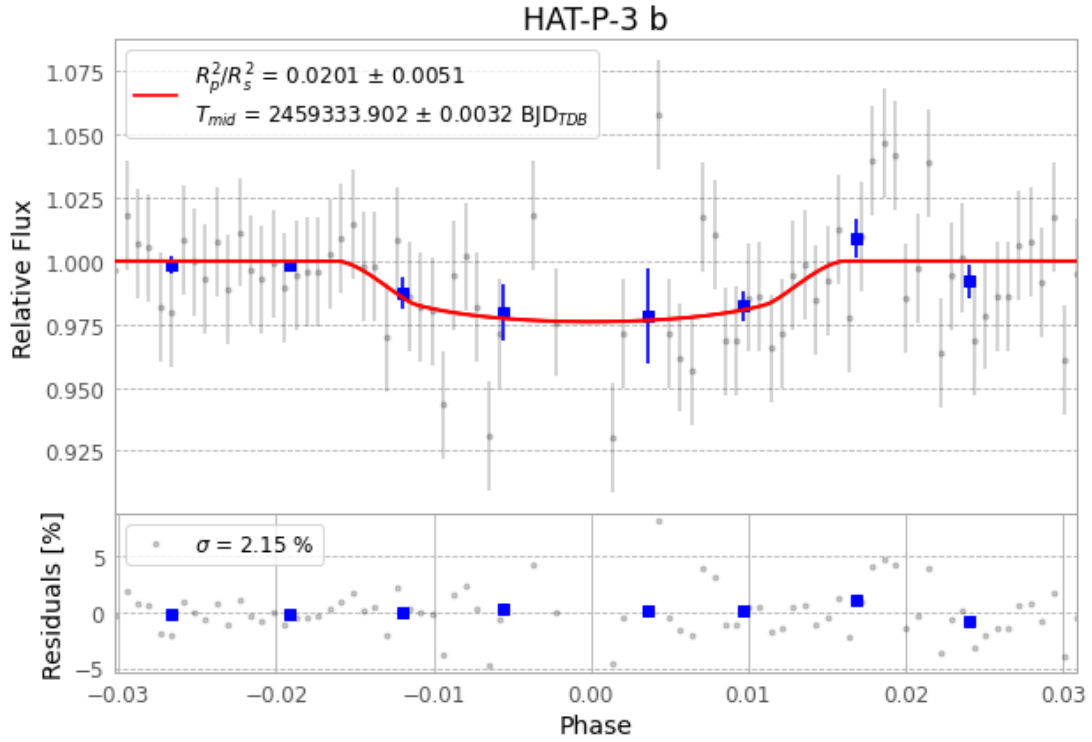


Figure 1 — FinalLightCurve_HAT-P-3 b_2021-04-29.png (SHA-256 c04e122a00a797bf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [414, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d14e917c784662ac...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), HATP-3210429072712.FITS → HATP-3210429114212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-210429.log (82ea3ef5e813...), FinalLightCurve_HAT-P-3 b_2021-04-29.png (c04e122a00a7...), FinalLightCurve_HAT-P-3 b_2021-04-29.csv (fb5d0df78f06...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 12:12Z.